

VAE: Variational Autoencoder:

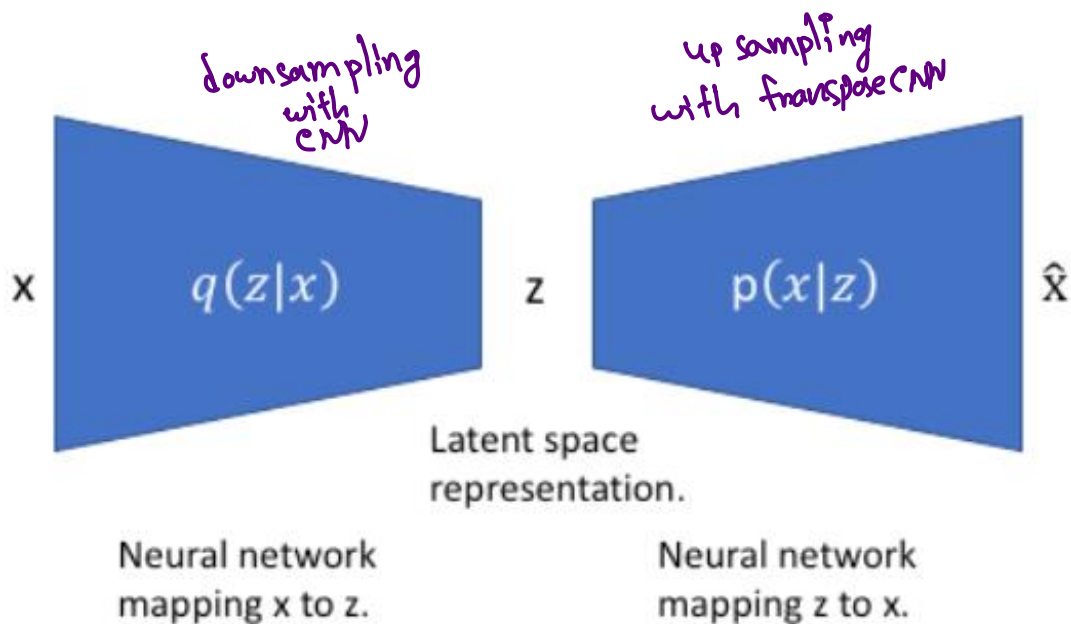
<https://lilianweng.github.io/posts/2018-08-12-vae/#vq-vae-and-vq-vae-2>

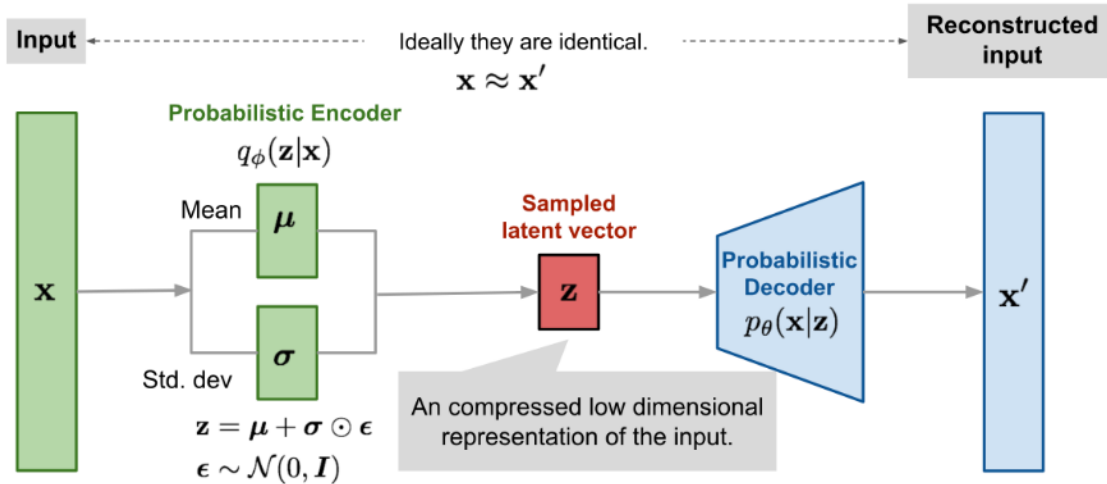
<https://www.jeremyjordan.me/variational-autoencoders/>

<https://www.youtube.com/watch?v=qJeaCHQ1k2w>



- Objective/ Goal of VAE: Trying to find distribution $q_{\phi}(z|x)$ from where we sample $z \sim q_{\phi}(z|x)$ to generate new sample x' from $p_{\theta}(x|z)$





$$L_{\text{VAE}}(\theta, \phi) = -\mathbb{E}_{z \sim q_{\phi}(z|x)} \log p_{\theta}(x|z) + D_{\text{KL}}(q_{\phi}(z|x) || p_{\theta}(z))$$

2. Architecture of VAE

- Encoder $q_{\phi}(z|x)$:
 - Maps the input x to a latent distribution $q_{\phi}(z|x)$, which is typically modeled as a Gaussian distribution.
- Latent Variable Sampling $z \sim q_{\phi}$:
 - z represents compressed information from x .
 - Sampling allows z to capture variability in the data.
- Decoder $p_{\theta}(x|z)$:
 - Reconstructs or generates samples x' based on the latent variable z .

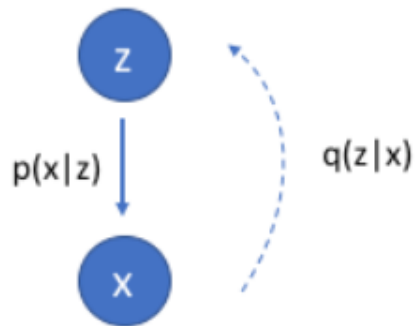
→ $q_{\phi}(z|x)$ ও সাধারণত $\mathcal{P}(\mu, \sigma)$, Normal distribution তে তুলনা করছি। অর্থাৎ $\mu=1, \sigma=0$ হিসাব করে নিয়েছি। আসলেই মনে হবে।

3. Learning Process

- Optimization Variables:
 - ϕ : Parameters of the encoder (probabilistic mapping to latent space).
 - θ : Parameters of the decoder (probabilistic mapping from latent space back to the data).
- Both are learned through **backpropagation** using the VAE loss function.

Loss Function:

- Balances two terms:
 - Reconstruction Loss:** Ensures x' is similar to x (e.g., log-likelihood).
 - KL Divergence:** Regularizes $q_{\phi}(z|x)$ to be close to a prior distribution (commonly standard Gaussian).



4. Graphical Model

- Highlights the probabilistic relationships:
 - z : Latent variable (hidden).
 - x : Observed variable (input).
 - Arrows show how z is sampled using $q_\phi(z|x)$, and x' is generated using $p_\theta(x|z)$.

We'd like to use our observations to understand the hidden variable.

✓ 5. Steps Summarized:

- Input x is fed into the encoder to compute $q_\phi(z|x)$.
- Sample z from $q_\phi(z|x)$.
- Use z as input to the decoder $p_\theta(x|z)$ to generate x' .
- Optimize the loss function to align x' with x while regularizing $q_\phi(z|x)$.

1. Goal of VAE

- The objective of VAEs is to learn a **distribution $q_\phi(z|x)$** of some **latent variables z** . From which we can sample **$z \sim q_\phi(z|x)$ to generate x' from $p_\theta(x|z)$**
- Instead of just encoding and decoding data deterministically (as in typical autoencoders), VAEs aim to:
 - Encode** input x into a latent space z such that z represents **a probability distribution (usually Gaussian)**.
 - Generate** new samples by sampling from this latent space distribution **$z \sim q_\phi(z|x)$** .

- Use a decoder $p_\theta(x|z)$ to reconstruct samples x' .

✓ Expectation of a random variable

(কম্পিউটার expectation বোঝে কঠোর নয়,
just x is কমান্ডিং $f(x)$ দিচ্ছি দিচ্ছি)

continuous random variable

$$E_x[f(x)] = \int f(x) p(x) dx$$

weighted avg.
discrete PDFs
roll a die: $\sum_{i=1}^n x_i * P(x_i)$

✓ Chain rule of probability

$$P(x, y) = P(x|y)P(y)$$

✓ Bayes' Theorem

$$P(x | y) = \frac{P(y|x)P(x)}{P(y)}$$

Kullback-Leibler Divergence

$$D_{KL}(P||Q) = \int p(x) \log \left(\frac{p(x)}{q(x)} \right) dx$$

✓ Properties:

- Not symmetric.
- Always ≥ 0
- It is equal to 0 if and only if $P = Q$

$$X \in [0, 10]$$

Random Variable

$$x \sim X$$

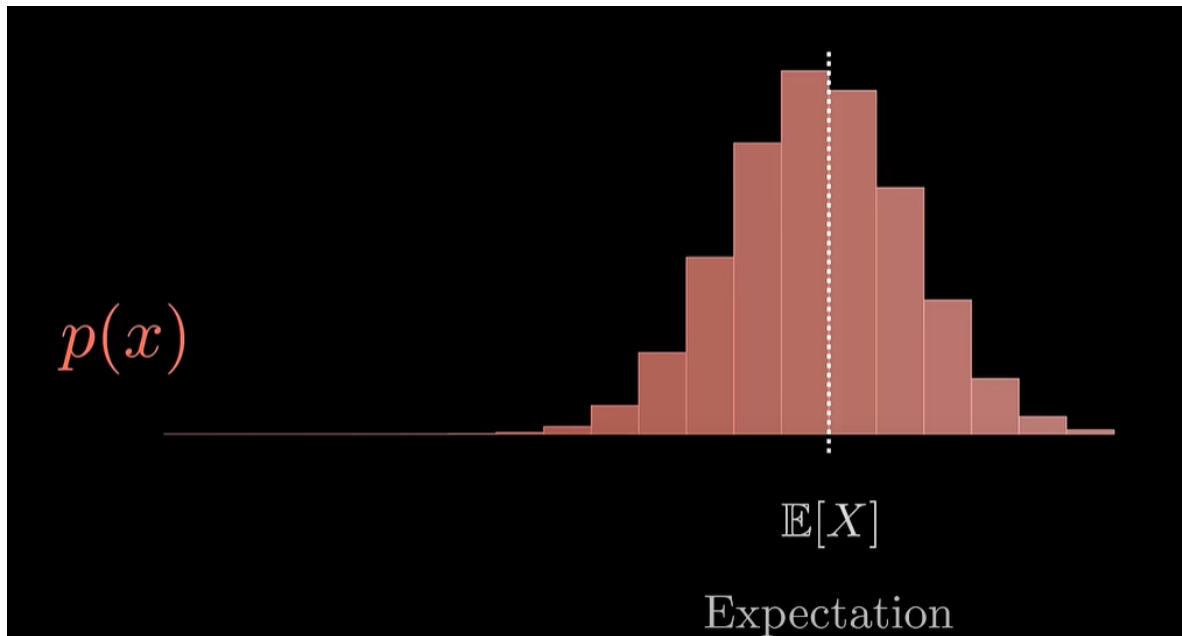
Sampling

$p(x)$

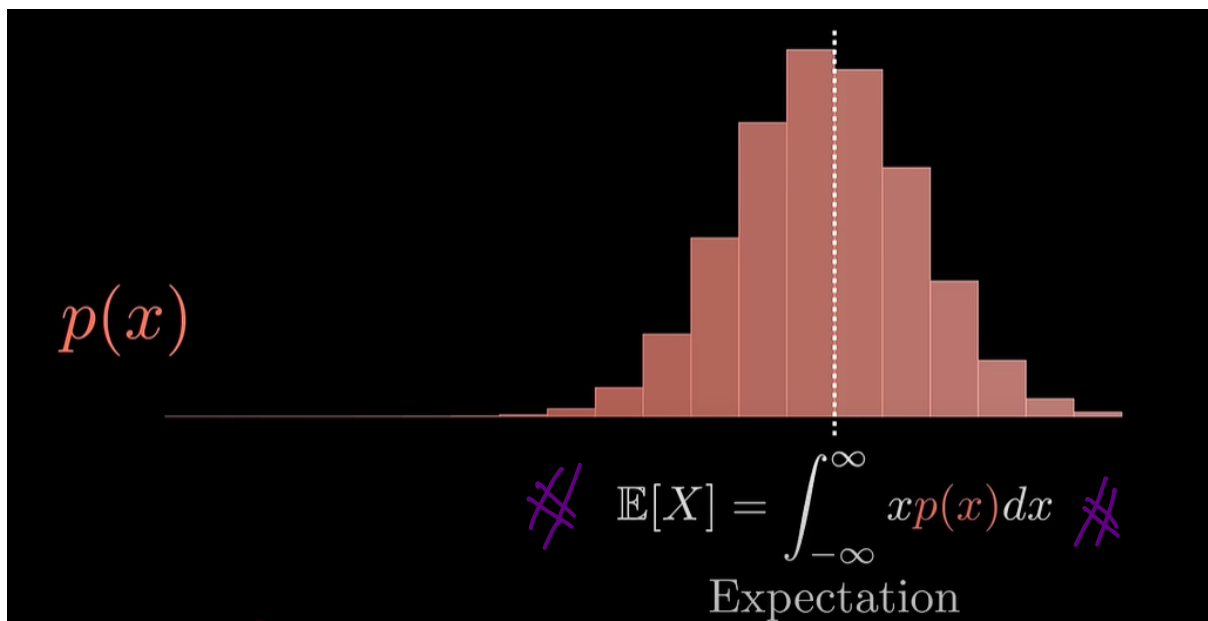
Probability Density Function

✓

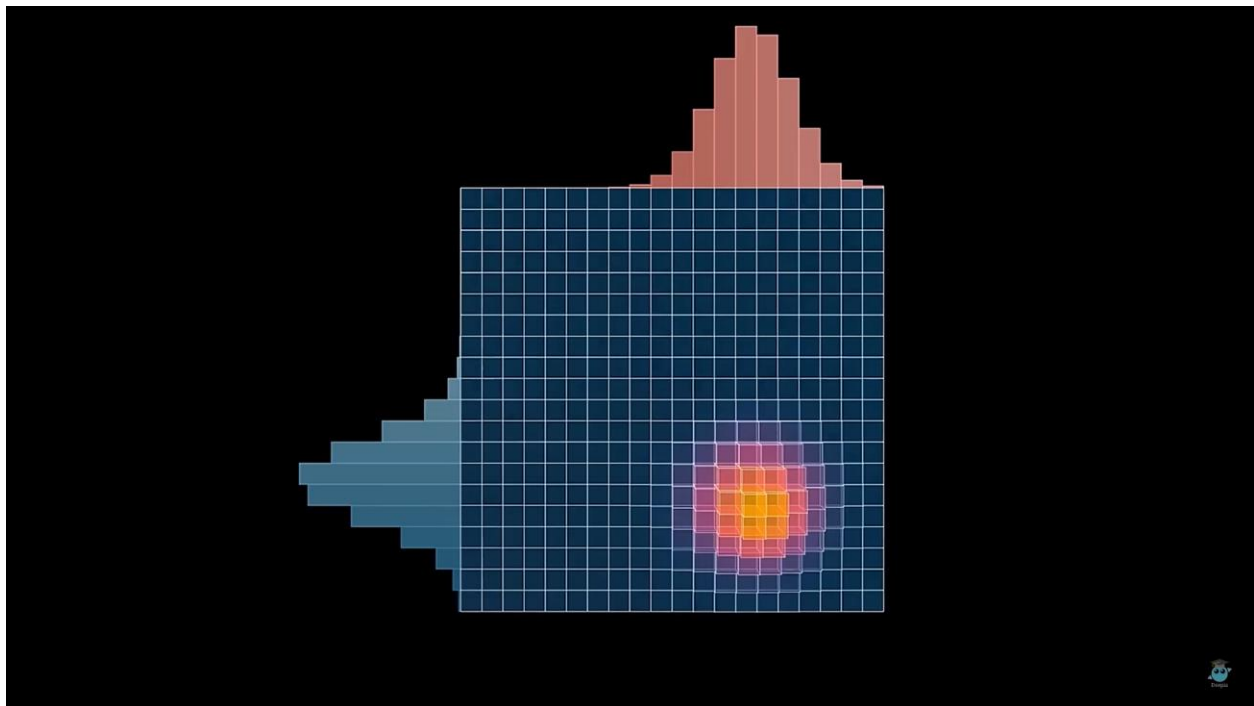
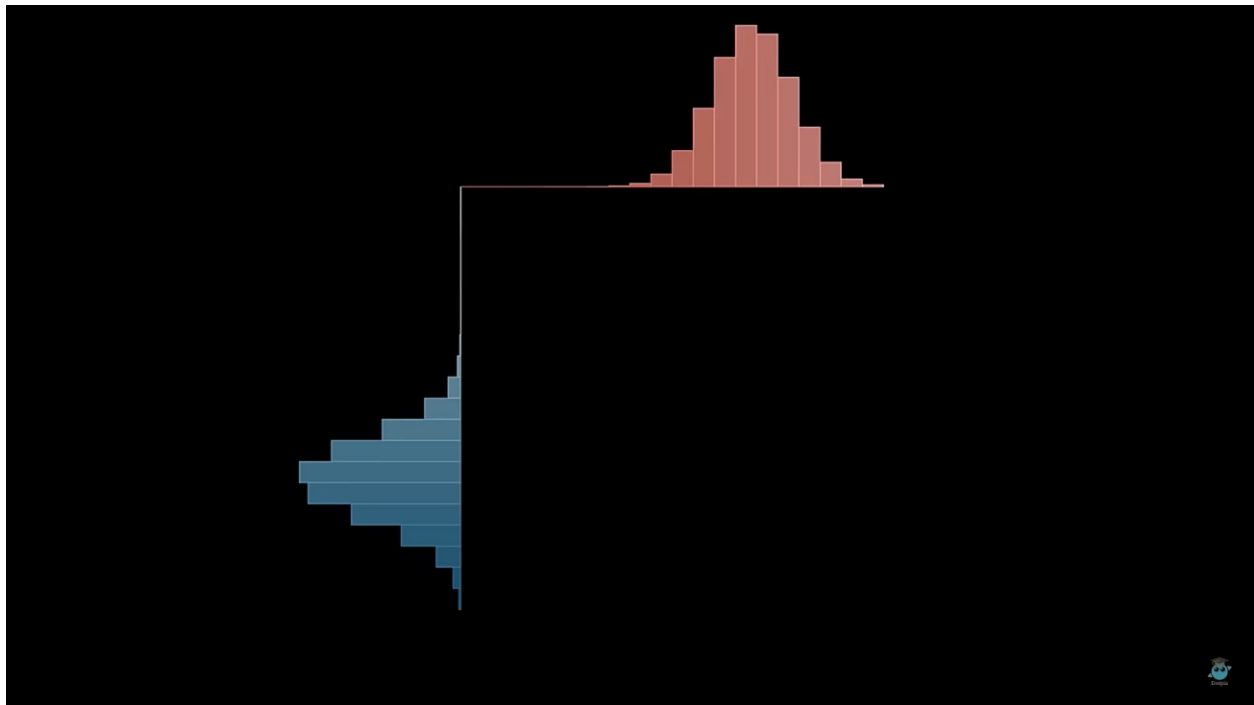
P(x) is the probability of any value of X



✓ (The interesting quantity is $\mathbb{E}[X]$ is the expectation of X ,. It the average value of X when we sampling from X . and is computed by the following formulation. CAN show the previous Lecture example



✓✗✗ Let consider two random variable X and Z , and their joint probability



In 3D, we can see the joint probability of X and Z. it shows **the each pair of events occurs together.**

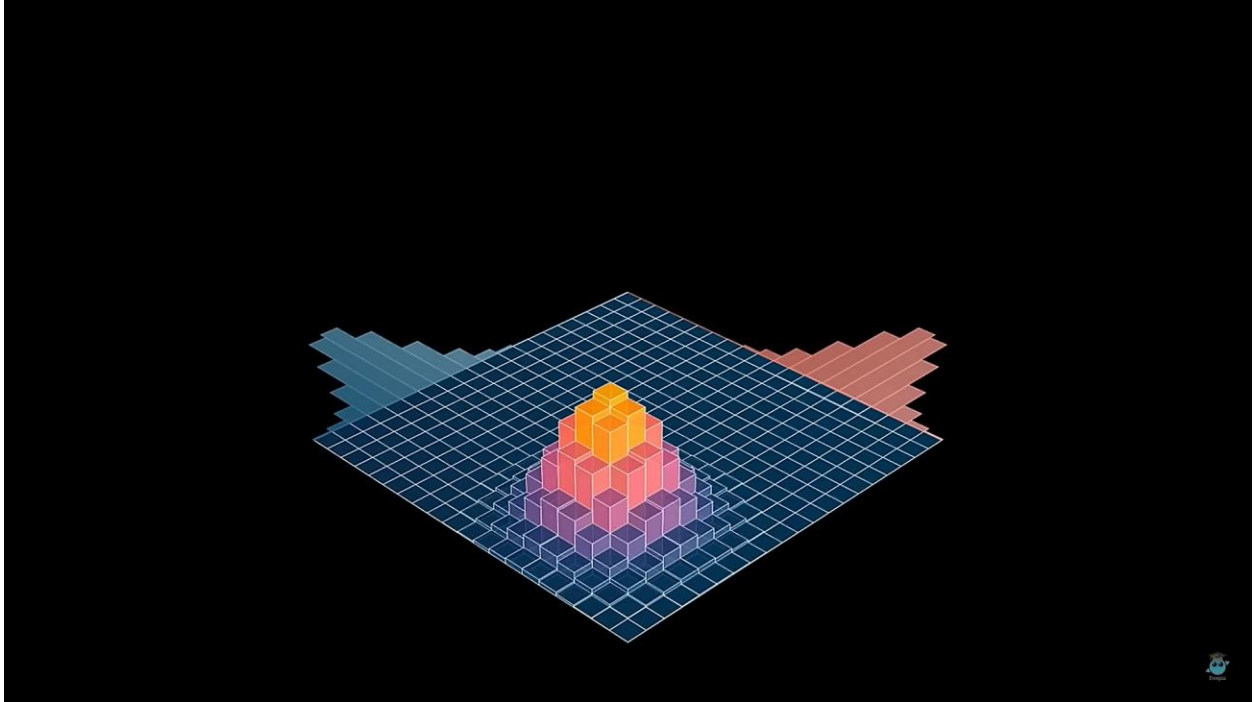
[show examples of joint distribution.]

If $X = \{H, T\}$, and $Z = \{1, 2, 3, 4, 5, 6\}$

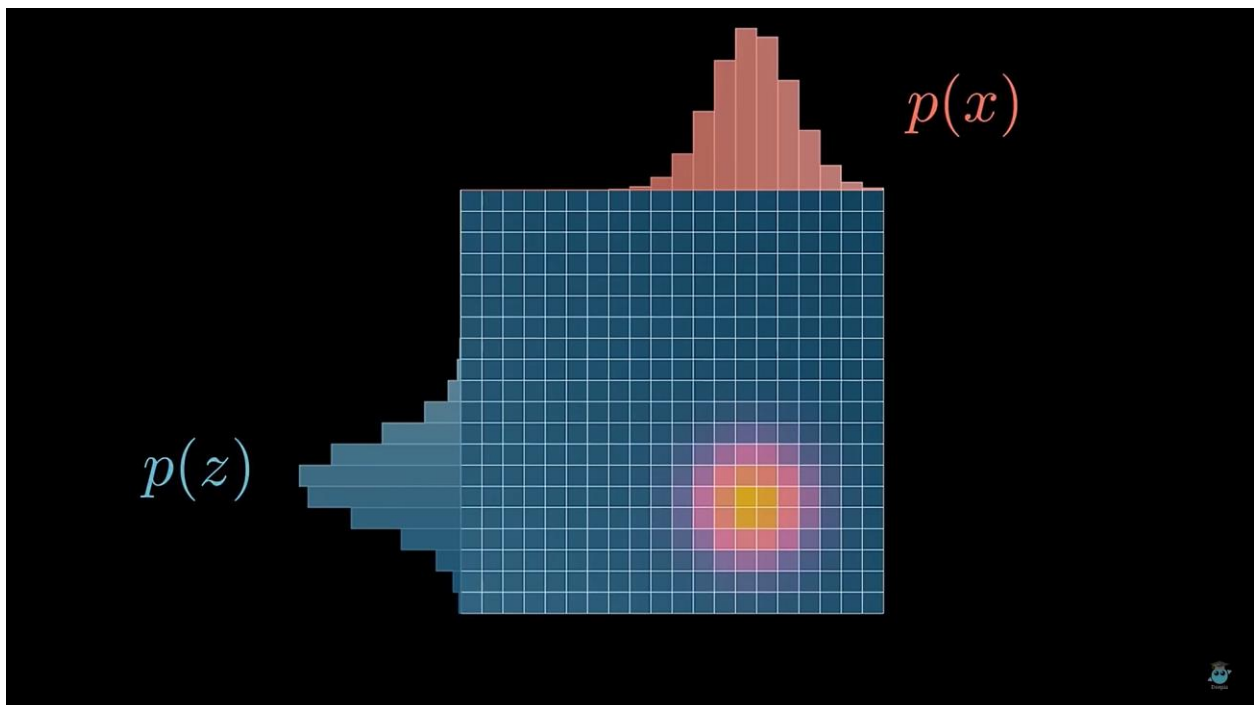
$$P(X = H, Y = 3) = 1/2 * 1/6 = 1/12$$

→ Probability distribution
when $X = \text{Head}$, $Y = 3$
Independent event are,
 $P(A \cap B) = P(A) * P(B)$

→ Example, continuous random variable
are, Example 4 discrete
random variable are. like coin
toss.



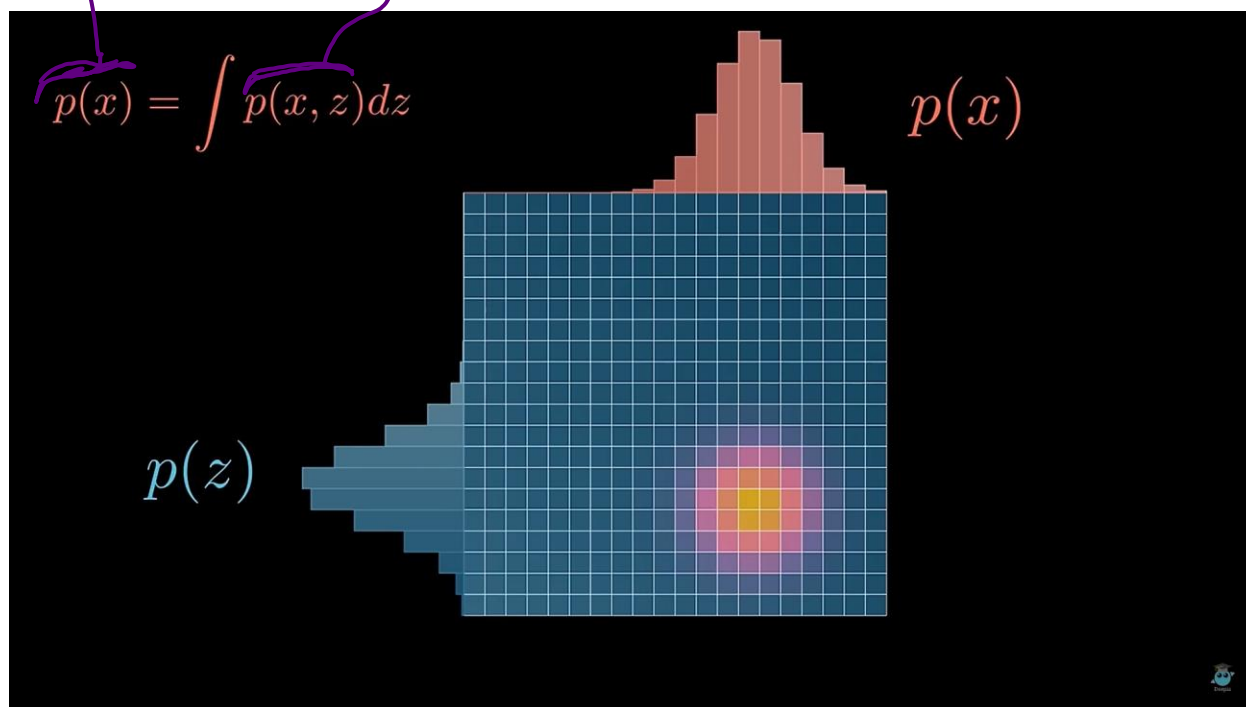
* (At the same time each Random variable has its own probability distribution called the marginal distribution.)



** Interesting property of joint distribution is to find $p(x)$ and $p(z)$ called the marginal distribution.

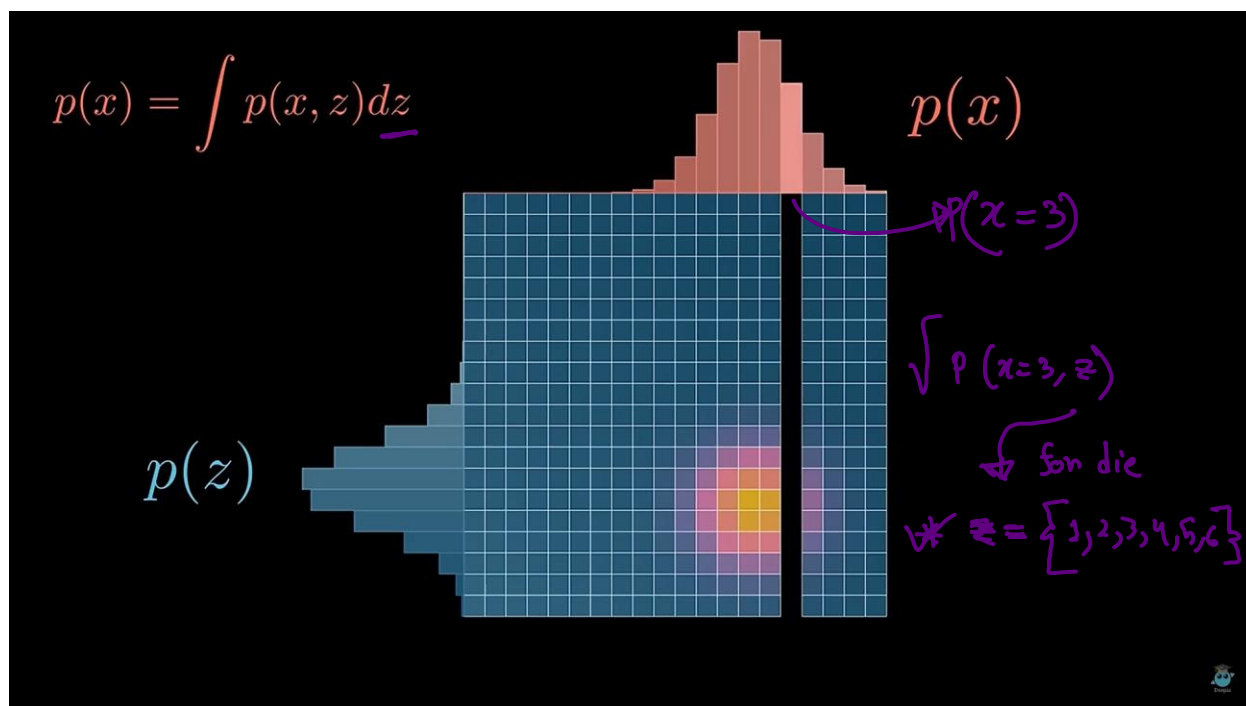
স্মারক, আমরা এই joint probability observe করছি, যা এই joint probability ত
এই কোন এক random variable এর probability দেখে বোঝে যা
marginal distribution. ✖ (Bayse Theorem)

Marginal probability → Joint Probability



✓ Example define the find the marginal distribution p(x) with respect to other variable z by integrating the joint distribution of p(x, z) with respect to dz

P(x=3) = find the joint probability over the all value of z



[show examples of joint distribution.]

$P(X) = \text{sum the joint probability over all the possible value of } Z.$

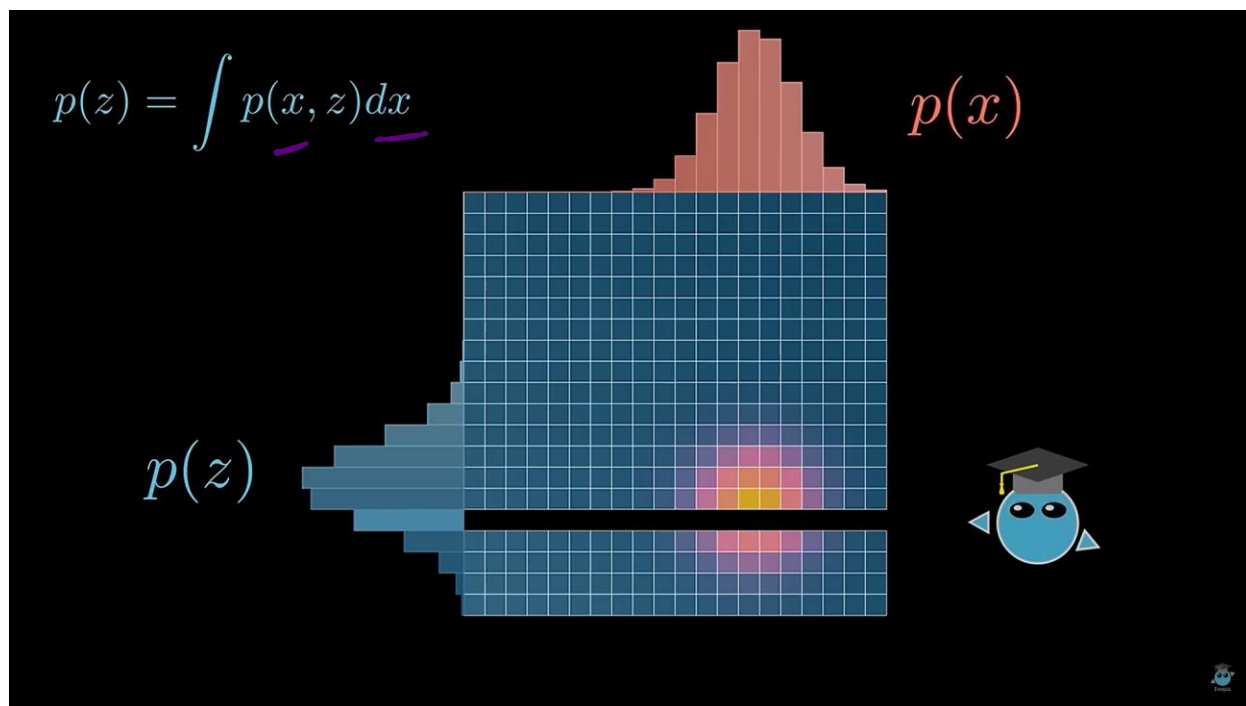
If $X = \{H, T\}$, and $Z = \{1, 2, 3, 4, 5, 6\}$

$$p(x) = \int p(x, z) dz$$

X (independent event)

$$P(X) = P(x=H) = P(X=H) * P(Y=1) + P(X=H) * P(Y=2) + P(X=H) * P(Y=3) + P(X=H) * P(Y=6) = P(X=H) * P(Y=2)$$

$$P(X) = \frac{1}{2} * \frac{1}{6} + \frac{1}{2} * \frac{1}{6} + \frac{1}{2} * \frac{1}{6} + \frac{1}{2} * \frac{1}{6} + \frac{1}{2} * \frac{1}{6} + \frac{1}{2} * \frac{1}{6} = 6 * \frac{1}{12} = \frac{1}{2}$$



$P(Z) = \text{sum the joint probability over all the possible value of } x.$

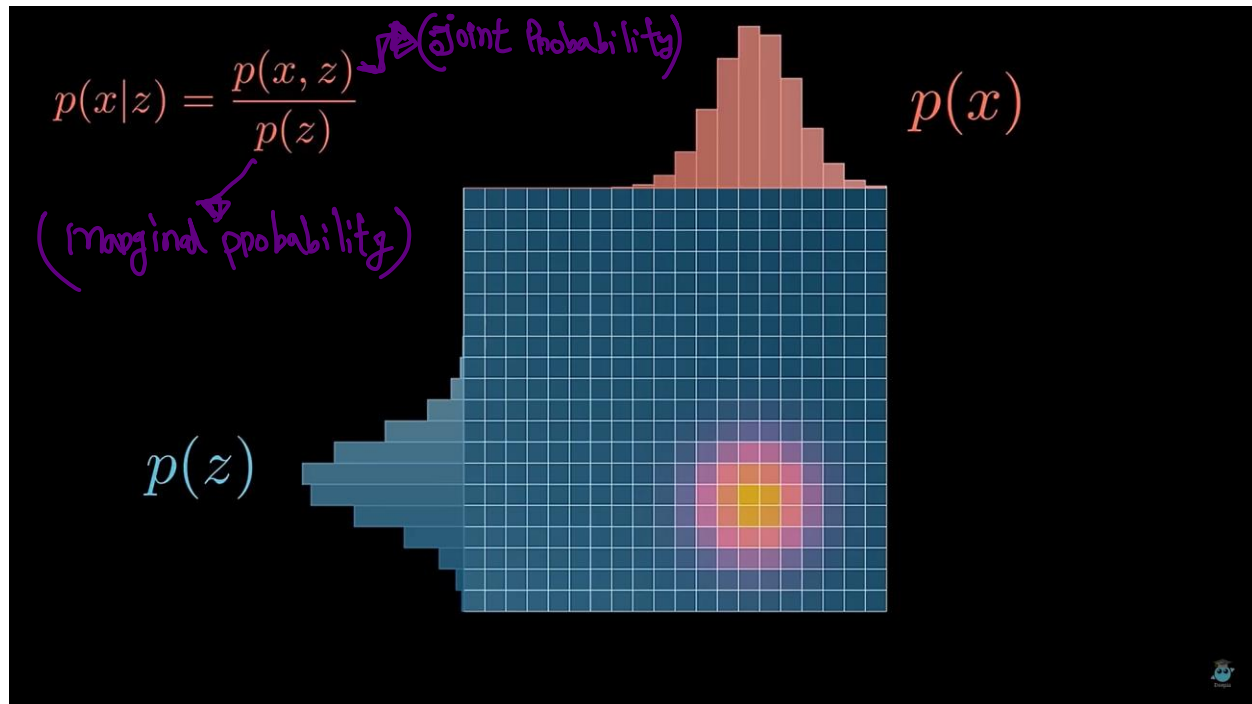
If $X = \{H, T\}$, and $Z = \{1, 2, 3, 4, 5, 6\}$

$$P(Z) = P(z=1) = P(X=H) * P(z=1) + P(X=T) * P(z=1) = \frac{1}{2} * \frac{1}{6} + \frac{1}{2} * \frac{1}{6} = \frac{1}{6}$$

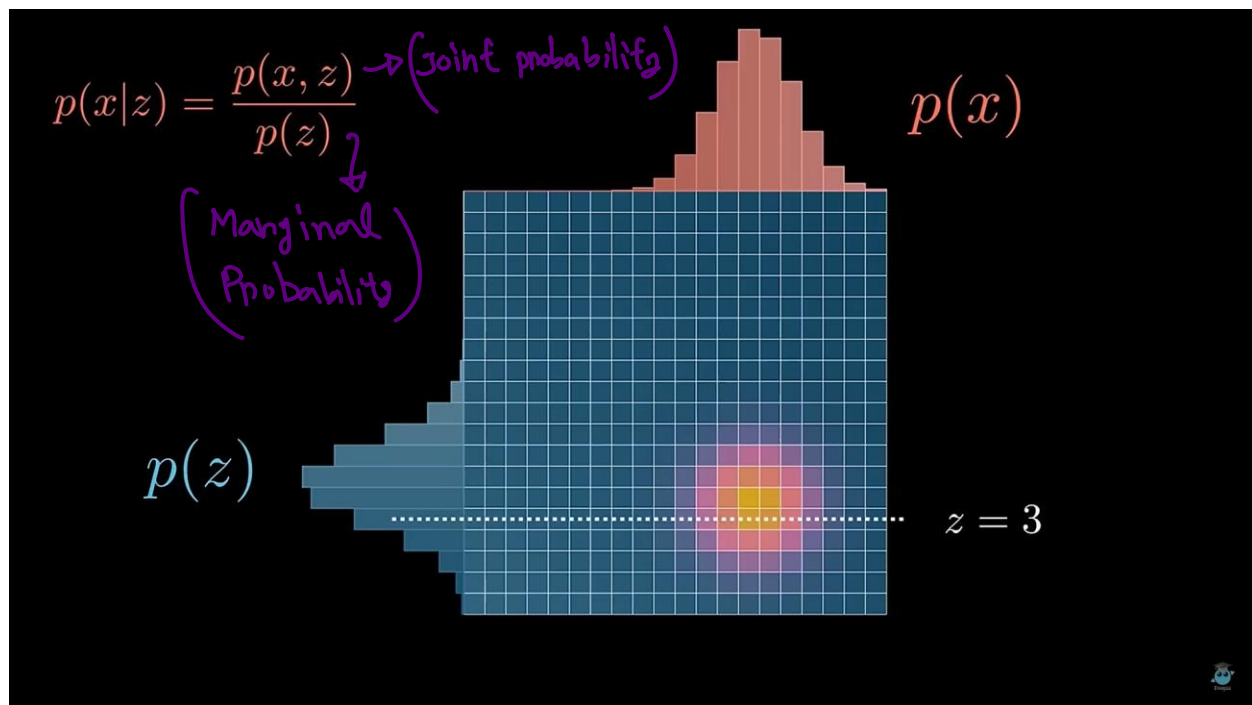
$$P(X=H, z=1) + P(X=T, z=1) = P(X=H) * P(z=1) + P(X=T) * P(z=1)$$

$$= \left(\frac{1}{2} * \frac{1}{6} \right) + \left(\frac{1}{2} * \frac{1}{6} \right) = \frac{1}{6}$$

$\propto$ (**Conditional probability**) it is defined as the joint distribution and normalizing it with the marginal distribution.



It is the slice of joint distribution and normalized it with the marginal distribution.



(Z=1)

It is slice of joint probability for fix $z = 3$, p dividing

$P(Z)$ = sum the joint probability over all the possible value of x .

If $X = \{H, T\}$, and $Z = \{1, 2, 3, 4, 5, 6\}$

$$P(Z) = P(z=1) = P(X=H) * P(z=1) + P(X=T) * P(z=1) = \frac{1}{2} * \frac{1}{6} + \frac{1}{2} * \frac{1}{6} = \frac{1}{6}$$

$$\text{As } P(x/z=1) = \frac{p(x, z=1)}{p(z=1)}$$

$$p(x=H, z=1) = p(x=H) * p(z=1) = \frac{1}{2} * \frac{1}{6} = \frac{1}{12}$$

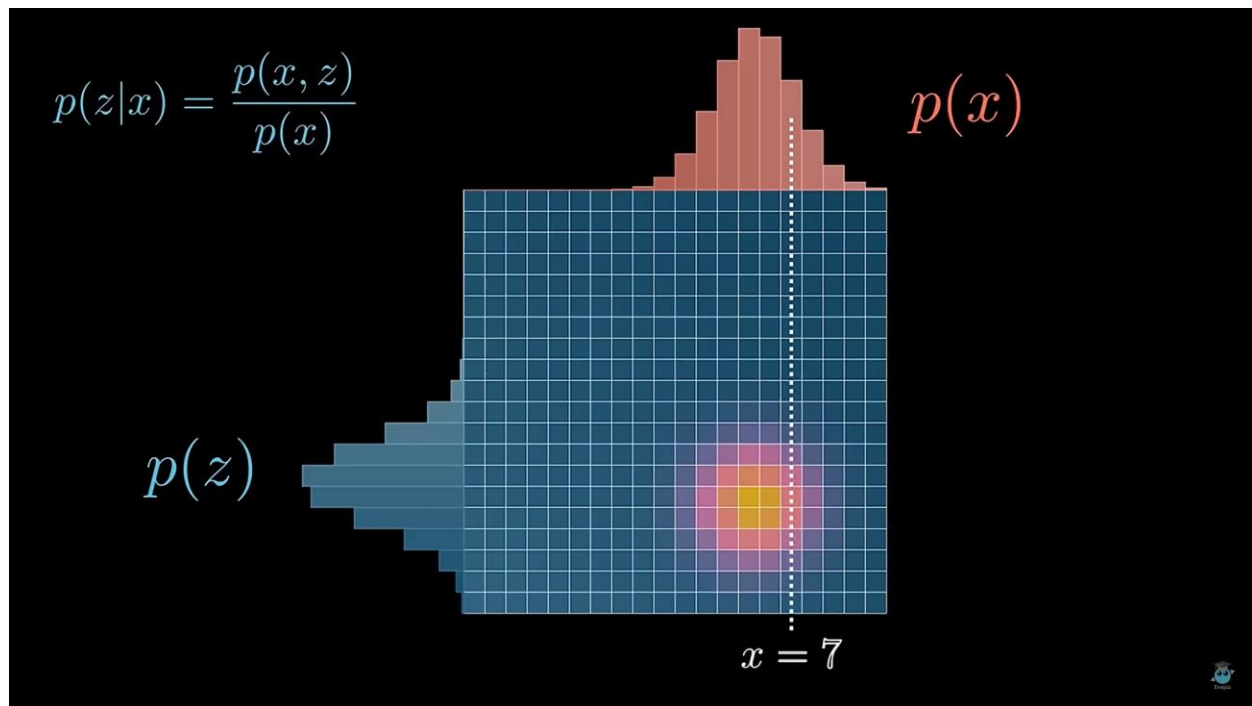
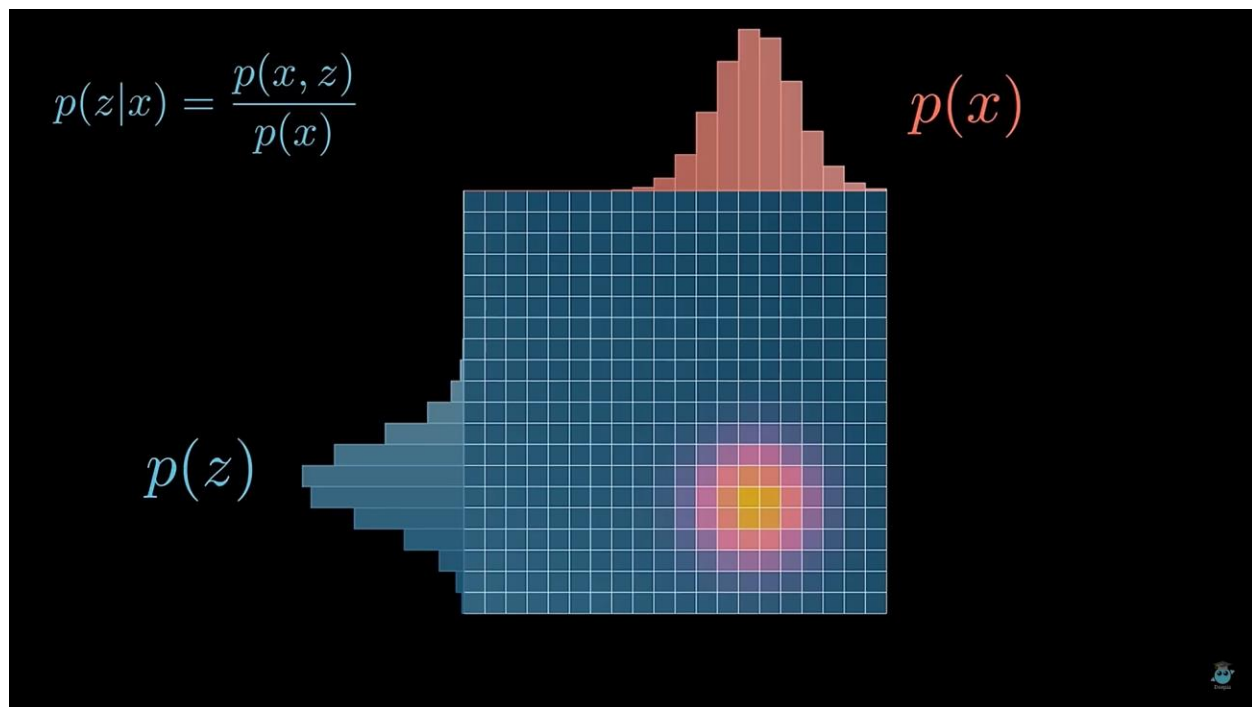
$$p(x=T, z=1) = p(x=T) * p(z=1) = \frac{1}{2} * \frac{1}{6} = \frac{1}{12}$$

therefore

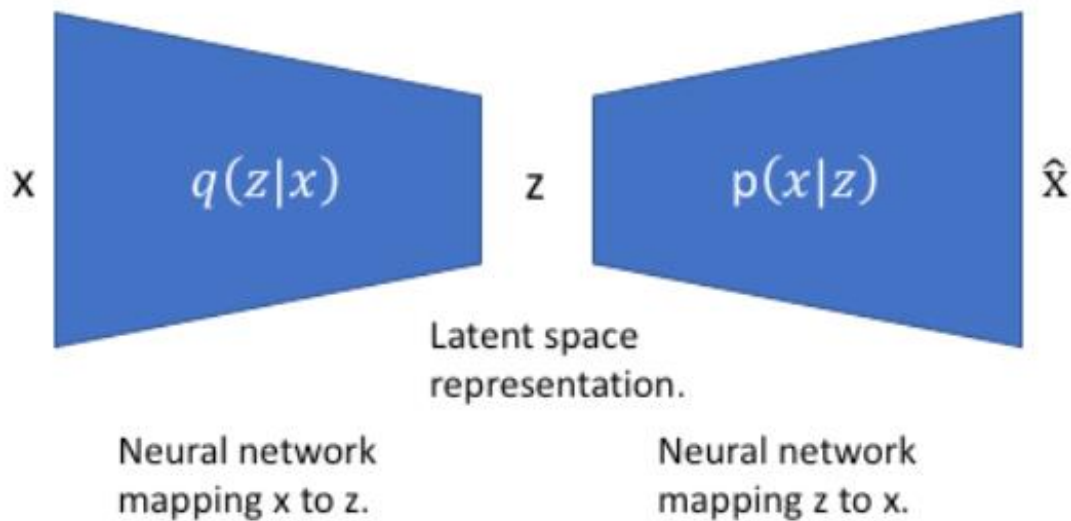
$$p(x=H/Z=1) = \frac{p(x=H, z=1)}{p(z=1)} = \frac{1}{12} * 6 = \frac{1}{2}$$

$$p(x=T/Z=1) = \frac{p(x=T, z=1)}{p(z=1)} = \frac{1}{12} * 6 = \frac{1}{2}$$

This makes sense because the coin toss is independent of the die roll, and thus the outcomes of the coin toss (H or T) remain equally likely even when we know $Z=3$.



Show the presence of z when we sample the data from x .



In the case of a Variational Autoencoder (VAE), **directly finding $P(Z|X)$ using the formula $P(Z|X) = P(X, Z) / P(X)$** is extremely difficult due to several reasons:

Why $P(Z|X) = \frac{P(X, Z)}{P(X)}$ Is Hard to Compute in VAEs

✓ 1. Marginal Probability $P(X)$ is Intractable:

- The denominator $P(X)$ requires integrating over all possible values of the latent variable Z , i.e.:

$$P(X) = \int P(X, Z) dZ.$$

- If Z is high-dimensional (e.g., a 10-dimensional latent space), the integration becomes computationally infeasible due to the curse of dimensionality.

উপরে, $\int$ এর মান ২ টা দিই, কিন্তু
 (খালি $\int$ " " ২ টা থাকবে না।
 হুঁ একটা মান মডেল করা থাকবে, আর সেটা জটিল
 integral
 করতে হবে।

✓ 2. Complex Form of $P(X, Z)$:

- In a VAE, $P(X, Z) = P(Z)P(X|Z)$, where:
 - $P(Z)$: The prior on the latent variable Z , typically a multivariate Gaussian.
 - $P(X|Z)$: The likelihood, often modeled by a complex neural network (the decoder in the VAE).
- Computing $P(X|Z)$ involves evaluating the decoder network for each possible value of Z , which is computationally expensive.

The Problem Setup:

1. We are trying to compute the marginal probability $P(X)$:

$$P(X) = \int P(X, Z) dZ = \int P(Z)P(X|Z) dZ.$$

2. Here:

- $P(X|Z)$: The likelihood of observing X given latent variables Z . This is usually a complex function, often modeled with a neural network.
- $P(Z)$: The prior over Z . In VAEs, this is typically a multivariate Gaussian distribution, e.g., $\mathcal{N}(0, I)$.
- dZ : The integration must consider all possible values of Z .

3. Dimension of Z :

- If Z is 2-dimensional ($Z = (Z_1, Z_2)$), then the integral becomes:

$$P(X) = \int_{Z_1} \int_{Z_2} P(Z_1, Z_2)P(X|Z_1, Z_2) dZ_1 dZ_2.$$

4. Why We Use Numerical Integration:

- For many models, especially in deep learning, the likelihood $P(X|Z)$ is not an analytically simple function, and closed-form solutions to this integral are not possible.
- Thus, we approximate the integral numerically by discretizing the continuous space of Z_1 and Z_2 into a grid.

How Grid-Based Numerical Integration Works

Discretizing Z_1 and Z_2 :

- Imagine Z_1 and Z_2 are continuous variables in the range $[a, b]$.
- To compute the integral, we divide the range of each variable into n discrete points:

$$\{z_1^{(1)}, z_1^{(2)}, \dots, z_1^{(n)}\} \quad \text{and} \quad \{z_2^{(1)}, z_2^{(2)}, \dots, z_2^{(n)}\}.$$

- Now, we evaluate the integrand $P(X|Z)P(Z)$ at each of the $n \times n$ combinations of (Z_1, Z_2) .

Numerical Approximation of the Integral:

- The integral is approximated as:

$$P(X) \approx \sum_{i=1}^n \sum_{j=1}^n P(X|Z_1^{(i)}, Z_2^{(j)})P(Z_1^{(i)}, Z_2^{(j)})\Delta Z_1 \Delta Z_2,$$

where ΔZ_1 and ΔZ_2 are the widths of the discretized intervals.

Computational Cost:

- For every combination of $Z_1^{(i)}$ and $Z_2^{(j)}$, we evaluate $P(X|Z)P(Z)$.
- The total number of evaluations required is $n \times n = O(n^2)$.

Toy Example

Suppose:

- Z_1 and Z_2 range from $[0, 1]$.
- Divide each into $n = 10$ points: $\{0.0, 0.1, 0.2, \dots, 1.0\}$.

To Compute $P(X)$:

- For each of the $n^2 = 10 \times 10 = 100$ grid points (Z_1, Z_2) , evaluate:
 - $P(X|Z_1, Z_2)$: Likelihood at each grid point.
 - $P(Z_1, Z_2)$: Prior at each grid point.
- Sum over all 100 combinations to approximate $P(X)$.

Scaling Issue:

- If Z were 10-dimensional $(Z_1, Z_2, \dots, Z_{10})$:
 - $n^{10} = 10^{10} = 10,000,000,000$ evaluations would be needed.

Problem Setup

Suppose we want to compute:

$$I = \int_0^1 f(z_1) dz_1,$$

where $f(z_1) = z_1^2 + 2z_1 + 1$.

Step-by-Step Solution

1. Divide the Range into Intervals

For $n = 5$, the interval length is:

$$\Delta z_1 = \frac{1}{n} = \frac{1}{5} = 0.2.$$

The interval midpoints for z_1 are:

$$z_1 = \{0.1, 0.3, 0.5, 0.7, 0.9\}.$$

2. Compute $f(z_1)$ at Each Midpoint

Evaluate $f(z_1)$ at each midpoint:

$$f(0.1) = 0.1^2 + 2(0.1) + 1 = 1.21,$$

$$f(0.3) = 0.3^2 + 2(0.3) + 1 = 1.69,$$

$$f(0.5) = 0.5^2 + 2(0.5) + 1 = 2.25,$$

$$f(0.7) = 0.7^2 + 2(0.7) + 1 = 2.89,$$

$$f(0.9) = 0.9^2 + 2(0.9) + 1 = 3.61.$$

3. Apply the Midpoint Rule

The midpoint rule approximates the integral as:

$$I \approx \Delta z_1 \cdot \sum_{i=1}^n f(z_1^{(i)}).$$

Substitute the values:

$$I \approx 0.2 \cdot (1.21 + 1.69 + 2.25 + 2.89 + 3.61).$$

4. Simplify the Calculation

$$I \approx 0.2 \cdot 11.65 = 2.33.$$

As $n=5$, there are 5 computation are needed inorder to solve the single integral, if $n=10$, then there are 10 computation are required to compute.

4. Toy Example to Compute Marginal Probability

Problem Setup:

- Let $Z = (z_1, z_2)$ be 2-dimensional latent variables, with both $z_1, z_2 \in [0, 1]$.
- Assume the prior distribution is uniform: $P(Z) = 1$ for $Z \in [0, 1]^2$.
- The likelihood $P(X|Z) = \exp(-(z_1 + z_2))$, representing some simple relationship between X and Z .
- The marginal $P(X)$ is:

$$P(X) = \int_0^1 \int_0^1 P(X|Z)P(Z) dz_1 dz_2 = \int_0^1 \int_0^1 \exp(-(z_1 + z_2)) dz_1 dz_2.$$

Solution:

Numerical integration for this 2D integral:

- Divide $[0, 1]$ for z_1 and z_2 into n points (e.g., $n = 10$).
- Sample grid points: $z_1, z_2 = [0, 0.1, 0.2, \dots, 1.0]$.
- Evaluate $f(z_1, z_2) = \exp(-(z_1 + z_2))$ at all n^2 grid points.

For $n = 10$:

- We compute $f(z_1, z_2)$ at $10 \times 10 = 100$ points.
- Sum up all values, weighted by the grid spacing ($\Delta z_1 \cdot \Delta z_2 = 0.1 \cdot 0.1 = 0.01$).

This process becomes increasingly expensive for higher dimensions.

Now we Z is two dimensional and if $n = 10$, then there are $10 \times 10 = 10^2$ computation are required to compute.

In practically Z is 10 to 100 dimensional, so it means 10^{10} to 10^{100} computation are required to compute, which is very expensive. So it seems intractable.

As the number of dimensions (or features/ $\mathbf{z}$'s) increases, the computational and data requirements grow exponential

<https://www.jeremyjordan.me/variational-autoencoders/>

Variational Inference:

Variational Inference (VI): Addresses the challenge of approximating the intractable posterior distribution $P_\theta(\mathbf{z}|\mathbf{x})$ by using a simpler, more manageable distribution $q_\phi(\mathbf{z}|\mathbf{x})$, often chosen as a Gaussian distribution.

The goal is to make $q_\phi(\mathbf{z}|\mathbf{x})$ closely approximate $P_\theta(\mathbf{z}|\mathbf{x})$. This is achieved by adjusting the parameters of $q_\phi(\mathbf{z}|\mathbf{x})$ (e.g., its mean and variance) to minimize the discrepancy between the two distributions.

To measure this discrepancy, the Kullback-Leibler (KL) divergence $KL(q_\phi(\mathbf{z}|\mathbf{x})||P_\theta(\mathbf{z}|\mathbf{x}))$ is minimized, effectively making $q_\phi(\mathbf{z}|\mathbf{x})$ as close as possible to $P_\theta(\mathbf{z}|\mathbf{x})$.

$$\begin{aligned}
 & D_{KL}(q_\phi(\mathbf{z}|\mathbf{x})||p_\theta(\mathbf{z}|\mathbf{x})) \\
 &= \int q_\phi(\mathbf{z}|\mathbf{x}) \log \frac{q_\phi(\mathbf{z}|\mathbf{x})}{p_\theta(\mathbf{z}|\mathbf{x})} d\mathbf{z} \\
 &= \int q_\phi(\mathbf{z}|\mathbf{x}) \log \frac{q_\phi(\mathbf{z}|\mathbf{x})p_\theta(\mathbf{x})}{p_\theta(\mathbf{z}, \mathbf{x})} d\mathbf{z} \\
 &= \int q_\phi(\mathbf{z}|\mathbf{x}) \left(\log p_\theta(\mathbf{x}) + \log \frac{q_\phi(\mathbf{z}|\mathbf{x})}{p_\theta(\mathbf{z}, \mathbf{x})} \right) d\mathbf{z} \\
 &= \log p_\theta(\mathbf{x}) + \int q_\phi(\mathbf{z}|\mathbf{x}) \log \frac{q_\phi(\mathbf{z}|\mathbf{x})}{p_\theta(\mathbf{z}, \mathbf{x})} d\mathbf{z} \\
 &= \log p_\theta(\mathbf{x}) + \int q_\phi(\mathbf{z}|\mathbf{x}) \log \frac{q_\phi(\mathbf{z}|\mathbf{x})}{p_\theta(\mathbf{x}|\mathbf{z})p_\theta(\mathbf{z})} d\mathbf{z} \\
 &= \log p_\theta(\mathbf{x}) + \mathbb{E}_{\mathbf{z} \sim q_\phi(\mathbf{z}|\mathbf{x})} \left[\log \frac{q_\phi(\mathbf{z}|\mathbf{x})}{p_\theta(\mathbf{z})} \right] - \log p_\theta(\mathbf{x}|\mathbf{z}) \\
 &= \log p_\theta(\mathbf{x}) + D_{KL}(q_\phi(\mathbf{z}|\mathbf{x})||p_\theta(\mathbf{z})) - \mathbb{E}_{\mathbf{z} \sim q_\phi(\mathbf{z}|\mathbf{x})} \log p_\theta(\mathbf{x}|\mathbf{z})
 \end{aligned}$$

(Formula of KL Divergence)

* [আপনার নোট করা অংশ থেকে জান দেখার।]
; Because $p(\mathbf{z}|\mathbf{x}) = p(\mathbf{z}, \mathbf{x})/p(\mathbf{x})$

$E_x[f(x)] = \int f(x) * p(x) dx$
; Because $\int q(\mathbf{z}|\mathbf{x}) d\mathbf{z} = 1$

; Because $p(\mathbf{z}, \mathbf{x}) = p(\mathbf{x}|\mathbf{z})p(\mathbf{z})$

Page 4 of this PDF

So we have: কম করা লাগবে। constant

$$D_{KL}(q_\phi(\mathbf{z}|\mathbf{x})||p_\theta(\mathbf{z}|\mathbf{x})) = \log p_\theta(\mathbf{x}) + D_{KL}(q_\phi(\mathbf{z}|\mathbf{x})||p_\theta(\mathbf{z})) - \mathbb{E}_{\mathbf{z} \sim q_\phi(\mathbf{z}|\mathbf{x})} \log p_\theta(\mathbf{x}|\mathbf{z})$$

*** constant term main aim করা কন্স E_z term কে বাড়ানো হবে। এর জন্য ELBO

Once rearrange the left and right hand side of the equation,

$$\log p_\theta(\mathbf{x}) - D_{KL}(q_\phi(\mathbf{z}|\mathbf{x})||p_\theta(\mathbf{z}|\mathbf{x})) = \mathbb{E}_{\mathbf{z} \sim q_\phi(\mathbf{z}|\mathbf{x})} \log p_\theta(\mathbf{x}|\mathbf{z}) - D_{KL}(q_\phi(\mathbf{z}|\mathbf{x})||p_\theta(\mathbf{z}))$$

The LHS of the equation is exactly what we want to maximize when learning the true distributions: we want to maximize the (log-)likelihood of generating real data (that is $\log p_\theta(\mathbf{x})$) and also minimize the difference between the real and estimated posterior distributions (the term D_{KL} works like a regularizer). Note that $p_\theta(\mathbf{x})$ is fixed with respect to q_ϕ .

$P(\mathbf{x}) = \frac{P(\mathbf{x}|\mathbf{z})P(\mathbf{z})}{P(\mathbf{z})}$
 $P(\mathbf{x}|\mathbf{z})$ ছাড়া হবে
 image generation অ জানা হবে।

↑ ELBO = (Reconst-term)↑ - (KL)↓ $\rightarrow$ यदि ELBO बढ़ाया जाये, जहाँ output बढ़ाया रहे।

$$\downarrow \text{loss} = -(\text{ELBO}) = -(\text{Reconst-term}) + \text{KL} \\ = \text{MSE} + \text{KL} \text{ (In code section)}$$

Instead of optimizing the LHS of the original loss function, we optimize the RHS, evidence lower bound (ELBO), which avoid the need to compute $P_\theta(\mathbf{z}|\mathbf{x})$

✓ The negation of the above defines our loss function:

$$\begin{aligned} L_{\text{VAE}}(\theta, \phi) &= -\log p_\theta(\mathbf{x}) + D_{\text{KL}}(q_\phi(\mathbf{z}|\mathbf{x})||p_\theta(\mathbf{z}|\mathbf{x})) \\ &= -\mathbb{E}_{\mathbf{z} \sim q_\phi(\mathbf{z}|\mathbf{x})} \log p_\theta(\mathbf{x}|\mathbf{z}) + D_{\text{KL}}(q_\phi(\mathbf{z}|\mathbf{x})||p_\theta(\mathbf{z})) \\ \theta^*, \phi^* &= \arg \min_{\theta, \phi} L_{\text{VAE}} \end{aligned}$$

In Variational Bayesian methods, this loss function is known as the *variational lower bound*, or *evidence lower bound*. The “lower bound” part in the name comes from the fact that KL divergence is always non-negative and thus $-L_{\text{VAE}}$ is the lower bound of $\log p_\theta(\mathbf{x})$.

$$-L_{\text{VAE}} = \log p_\theta(\mathbf{x}) - D_{\text{KL}}(q_\phi(\mathbf{z}|\mathbf{x})||p_\theta(\mathbf{z}|\mathbf{x})) \leq \log p_\theta(\mathbf{x})$$

Therefore by minimizing the loss, we are maximizing the lower bound of the probability of generating real data samples.

(23) $z = z \sim Q_q(z/x)$

Substituting (A) in (B)

$$\begin{aligned}
 D_{KL}(Q_q(z/x) \parallel P_0(z/x)) &= E_z [\log Q_q(z/x)] \\
 &\quad - E_z \left[\log \frac{P_0(x/z) P(z)}{P(x)} \right] \quad \text{Bayes' Theorem} \\
 &= E_z [\log Q_q(z/x)] - E_z \left[\log \frac{P(z, x)}{P(x)} \right] \\
 &= E_z [\log Q_q(z/x)] - E_z [\log P(z, x)] + E_z [\log P(x)] \\
 &= E_z [\log Q_q(z/x)] - E_z [\log P(z, x)] + \int Q_q(z/x) \log P(x) dz \\
 &= E_z [\log Q_q(z/x)] - E_z [\log P(z, x)] + \log P(x) \int Q_q(z/x) dz \\
 &= E_z [\log Q_q(z/x)] - E_z [\log P(z, x)] + \log P(x)
 \end{aligned}$$

$$\log P(x) = \underbrace{E_z [\log P(z, x)] - E_z [\log Q_q(z/x)]}_{\text{Component 1}} + \underbrace{D_{KL}(Q_q(z/x) \parallel P_0(z/x))}_{\text{Component 2}}$$

Component 1: $\uparrow$ max
 Component 2: $\downarrow$ min, we want to min. it

Fix no z

- In order to make the above equation equally true, if we minimize component (2) we have to maximize the component 1 (1).
- Component 1 is called Evidence Lower Bound (ELBO).
- Now if we maximize ELBO, it indirectly minimizes the KL divergence.

Q4)

• Why $\mathcal{L}$ is called lower bound

As $KL \geq 0$ (it is property of KL)

$$\therefore \log p_a(x) = \mathcal{L} + \underbrace{(\text{+ve no})}_{\geq 0}^{KL}$$

$$\mathcal{L} = \log p_a(x) - KL$$

• If $KL = 0$

$$\mathcal{L} = \log p_a(x)$$

• If $KL > 0$

$$\mathcal{L} < \log p_a(x)$$

$$\therefore \mathcal{L} \leq \log p_a(x)$$

So it is lower bound of $\log p_a(x)$

NOW As

$$\begin{aligned} \mathcal{L} = ELBO &= E_z [\log p(x/z)] - E_z [\log q_\phi(z/x)] \\ &= E_z [\log p(x/z) p_a(z)] - E_z [\log q_\phi(z/x)] \\ &= E_z [\log p(x/z)] + \underbrace{E_z [\log p_a(z)] - E_z [\log q_\phi(z/x)]}_{\geq 0} \\ &= E_z [\log p(x/z)] - E_z [\log \underbrace{q_\phi(z/x)}_{p_a(z)}] \end{aligned}$$

$$\mathcal{L} = E_z [\log p(x/z)] - \underbrace{D_{KL}[q_\phi(z/x) \parallel p_a(z)]}_{\geq 0}$$

expected reconstruction error

KL

if z is gaussian, then reconstruction is square error
 $p(x/z) = \frac{1}{\sigma^2} e^{-|x-z|^2/\sigma^2}$
 $\log p(x/z) = -|x-z|^2/\sigma^2$ if z is better, then reconstruction error is smaller

learned prior distribution

कुल ELBO Loss
बल, घण्टा skip करना

$$P(z|x) = \frac{P(x|z) P(z)}{P(x)}$$

$$\log p_{\theta}(x) = E_{q_{\varphi}(z|x)} \left[\log \frac{p_{\theta}(x, z)}{q_{\varphi}(z|x)} \right] + D_{KL} (q_{\varphi}(z|x) \| p_{\theta}(z|x))$$

ELBO = Evidence Lower Bound

ELBO

≥ 0

Total Compensation =

Base Salary

+

Bonus

≥ 0

We can for sure deduce the following:

Total Compensation $\geq$

Base Salary

$$\log p_{\theta}(x) \geq E_{q_{\varphi}(z|x)} \left[\log \frac{p_{\theta}(x, z)}{q_{\varphi}(z|x)} \right]$$